INCH-POUND

MIL-DTL-28840/20H
w/AMENDMENT 1
12 June 2019
SUPERSEDING
MIL-DTL-28840/20H
21 November 2013

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, CRIMP CONTACTS, RECEPTACLE, WALL MOUNTING ASSEMBLY, STRAIGHT, CLASSES DJ, DJS, PJ, TJ AND ZJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.

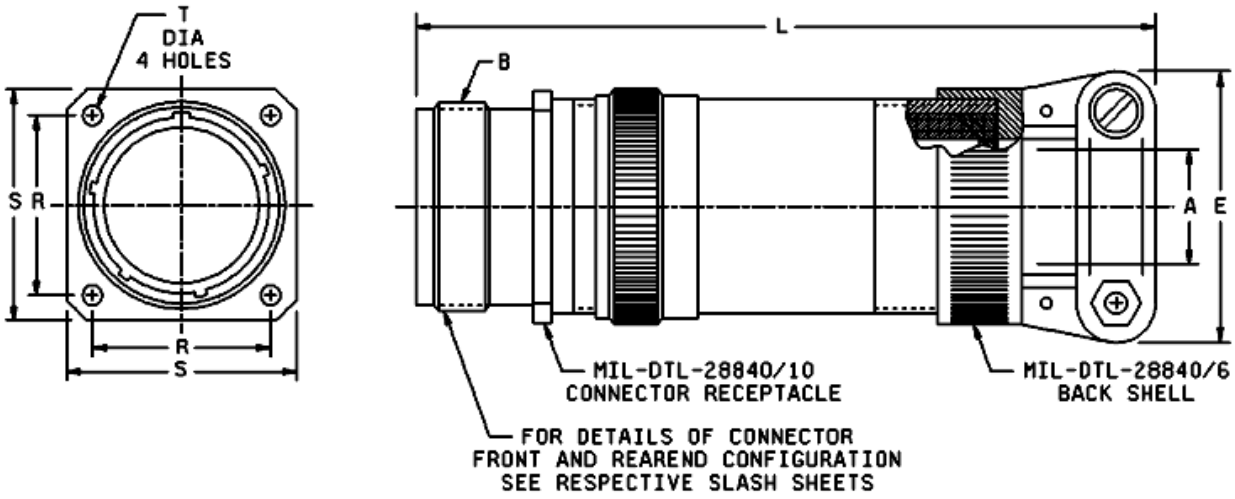


FIGURE 1. Dimensions and configuration.

AMSC N/A

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:41Z
Check the source to verify that this is the current version before use.

Shell size designator (see note 3)	Assembly Part or Identifying Number (PIN) (see note 6)	Connector receptacle (PIN) (see note 6)	Backshell assembly (PIN) (see note 6)	A Cable entry	
				max	min
A (11)	M28840/20*A*A***	M28840/10*A***	M28840/601**	.312 (7.92)	.188 (4.78)
A (11)	M28840/20*A*B***	M28840/10*A***	M28840/624**	.438 (11.13)	.281 (7.14)
A (11)	M28840/20*A*C***	M28840/10*A***	M28840/625**	.562 (14.27)	.344 (8.74)
B (13)	M28840/20*B*A***	M28840/10*B***	M28840/602**	.312 (7.92)	.188 (4.78)
B (13)	M28840/20*B*B***	M28840/10*B***	M28840/603**	.438 (11.13)	.281 (7.14)
B (13)	M28840/20*B*C***	M28840/10*B***	M28840/626**	.562 (14.27)	.344 (8.74)
B (13)	M28840/20*B*D***	M28840/10*B***	M28840/627**	.625 (15.88)	.375 (9.53)
C (15)	M28840/20*C*A***	M28840/10*C***	M28840/605**	.438 (11.13)	.281 (7.14)
C (15)	M28840/20*C*B***	M28840/10*C***	M28840/606**	.562 (14.27)	.344 (8.74)
C (15)	M28840/20*C*C***	M28840/10*C***	M28840/628**	.625 (15.88)	.375 (9.53)
C (15)	M28840/20*C*D***	M28840/10*C***	M28840/607**	.750 (19.05)	.438 (11.13)
C (15)	M28840/20*C*E***	M28840/10*C***	M28840/604**	.312 (7.92)	.188 (4.78)
D (17)	M28840/20*D*A***	M28840/10*D***	M28840/608**	.625 (15.88)	.375 (9.53)
D (17)	M28840/20*D*B***	M28840/10*D***	M28840/609**	.750 (19.05)	.438 (11.13)
D (17)	M28840/20*D*C***	M28840/10*D***	M28840/610**	.938 (23.83)	.625 (15.88)
E (19)	M28840/20*E*A***	M28840/10*E***	M28840/611**	.625 (15.88)	.375 (9.53)
E (19)	M28840/20*E*B***	M28840/10*E***	M28840/612**	.938 (23.83)	.625 (15.88)
F (23)	M28840/20*F*A***	M28840/10*F***	M28840/613**	.750 (19.05)	.438 (11.13)
F (23)	M28840/20*F*B***	M28840/10*F***	M28840/614**	.938 (23.83)	.625 (15.88)
F (23)	M28840/20*F*C***	M28840/10*F***	M28840/615**	1.188 (30.18)	.812 (20.62)

FIGURE 1. Dimensions and configuration – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:41Z
Check the source to verify that this is the current version before use.

Shell size designator (see note 3)	Assembly (PIN) (see note 6)	Connector receptacle (PIN) (see note 6)	Backshell assembly (PIN) (see note 6)	B thread class 2B	E ref	L max
A (11)	M28840/20*A*A***	M28840/10*A***	M28840/601**	.750-.1P.2L-D.S.	.938 (23.83)	6.625 (168.28)
A (11)	M28840/20*A*B***	M28840/10*A***	M28840/624**	.750-.1P.2L-D.S.	1.125 (28.58)	6.625 (168.28)
A (11)	M28840/20*A*C***	M28840/10*A***	M28840/625**	.750-.1P.2L-D.S.	1.312 (33.32)	6.625 (168.28)
B (13)	M28840/20*B*A***	M28840/10*B***	M28840/602**	.875-.1P.2L-D.S.	.938 (23.83)	6.625 (168.28)
B (13)	M28840/20*B*B***	M28840/10*B***	M28840/603**	.875-.1P.2L-D.S.	1.125 (28.58)	6.625 (168.28)
B (13)	M28840/20*B*C***	M28840/10*B***	M28840/626**	.875-.1P.2L-D.S.	1.312 (33.32)	6.625 (168.28)
B (13)	M28840/20*B*D***	M28840/10*B***	M28840/627**	.875-.1P.2L-D.S.	1.312 (33.32)	7.125 (180.98)
C (15)	M28840/20*C*A***	M28840/10*C***	M28840/605**	1.062-.1P.2L-D.S.	1.125 (28.58)	6.625 (168.28)
C (15)	M28840/20*C*B***	M28840/10*C***	M28840/606**	1.062-.1P.2L-D.S.	1.312 (33.32)	6.625 (168.28)
C (15)	M28840/20*C*C***	M28840/10*C***	M28840/628**	1.062-.1P.2L-D.S.	1.312 (33.32)	7.125 (180.98)
C (15)	M28840/20*C*D***	M28840/10*C***	M28840/607**	1.062-.1P.2L-D.S.	1.593 (40.46)	7.188 (182.58)
C (15)	M28840/20*C*E***	M28840/10*C***	M28840/604**	1.062-.1P.2L-D.S.	.938 (23.83)	6.625 (168.28)
D (17)	M28840/20*D*A***	M28840/10*D***	M28840/608**	1.125-.1P.2L-D.S.	1.312 (33.32)	7.125 (180.98)
D (17)	M28840/20*D*B***	M28840/10*D***	M28840/609**	1.125-.1P.2L-D.S.	1.593 (40.46)	7.188 (182.58)
D (17)	M28840/20*D*C***	M28840/10*D***	M28840/610**	1.125-.1P.2L-D.S.	1.750 (44.45)	7.188 (182.58)
E (19)	M28840/20*E*A***	M28840/10*E***	M28840/611**	1.312-.1P.2L-D.S.	1.312 (33.32)	7.125 (180.98)
E (19)	M28840/20*E*B***	M28840/10*E***	M28840/612**	1.312-.1P.2L-D.S.	1.750 (44.45)	7.250 (184.15)
F (23)	M28840/20*F*A***	M28840/10*F***	M28840/613**	1.500-.1P.2L-D.S.	1.593 (40.46)	7.688 (195.28)
F (23)	M28840/20*F*B***	M28840/10*F***	M28840/614**	1.500-.1P.2L-D.S.	1.750 (44.45)	7.812 (198.42)
F (23)	M28840/20*F*C***	M28840/10*F***	M28840/615**	1.500-.1P.2L-D.S.	2.093 (53.16)	7.938 (201.63)

FIGURE 1. Dimensions and configuration – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:41Z
Check the source to verify that this is the current version before use.

Shell size designator (see note 3)	Assembly (PIN) (see note 6)	Connector receptacle (PIN) (see note 6)	Backshell assembly (PIN) (see note 6)	A Cable entry	
				max	min
F (23)	M28840/20*F*B***	M28840/10*F***	M28840/614**	.938 (23.83)	.625 (15.88)
F (23)	M28840/20*F*C***	M28840/10*F***	M28840/615**	1.188 (30.18)	.812 (20.62)
G (25)	M28840/20*G*A***	M28840/10*G***	M28840/616**	.938 (23.83)	.625 (15.88)
G (25)	M28840/20*G*B***	M28840/10*G***	M28840/617**	1.188 (30.18)	.812 (20.62)
G (25)	M28840/20*G*C***	M28840/10*G***	M28840/618**	1.250 (31.75)	.875 (22.23)
G (25)	M28840/20*G*D***	M28840/10*G***	M28840/619**	1.380 (35.05)	1.000 (25.40)
H (29)	M28840/20*H*A***	M28840/10*H***	M28840/620**	1.188 (30.18)	.812 (20.62)
H (29)	M28840/20*H*B***	M28840/10*H***	M28840/621**	1.380 (35.05)	1.000 (25.40)
J (33)	M28840/20*J*A***	M28840/10*J***	M28840/622**	1.380 (35.05)	1.000 (25.40)
J (33)	M28840/20*J*B***	M28840/10*J***	M28840/623**	1.625 (41.28)	1.250 (31.75)

Shell size designator (see note 3)	Assembly (PIN) (see note 6)	Connector receptacle (PIN) (see note 6)	Backshell assembly (PIN) (see note 6)	B thread class 2B	E ref	L max
F (23)	M28840/20*F*B***	M28840/10*F***	M28840/614**	1.500-.1P.2L-D.S.	1.750 (44.45)	7.812 (198.42)
F (23)	M28840/20*F*C***	M28840/10*F***	M28840/615**	1.500-.1P.2L-D.S.	2.093 (53.16)	7.938 (201.63)
G (25)	M28840/20*G*A***	M28840/10*G***	M28840/616**	1.625-.1P.2L-D.S.	1.750 (44.45)	7.750 (196.85)
G (25)	M28840/20*G*B***	M28840/10*G***	M28840/617**	1.625-.1P.2L-D.S.	2.093 (53.16)	8.438 (214.33)
G (25)	M28840/20*G*C***	M28840/10*G***	M28840/618**	1.625-.1P.2L-D.S.	2.093 (53.16)	8.438 (214.33)
G (25)	M28840/20*G*D***	M28840/10*G***	M28840/619**	1.625-.1P.2L-D.S.	2.343 (59.51)	8.500 (215.90)
H (29)	M28840/20*H*A***	M28840/10*H***	M28840/620**	1.812-.1P.2L-D.S.	2.093 (53.16)	8.438 (214.33)
H (29)	M28840/20*H*B***	M28840/10*H***	M28840/621**	1.812-.1P.2L-D.S.	2.343 (59.51)	8.500 (215.90)
J (33)	M28840/20*J*A***	M28840/10*J***	M28840/622**	2.000-.1P.2L-D.S.	2.343 (59.51)	8.500 (215.90)
J (33)	M28840/20*J*B***	M28840/10*J***	M28840/623**	2.000-.1P.2L-D.S.	2.750 (69.85)	8.750 (222.25)

FIGURE 1. Dimensions and configuration – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:41Z
Check the source to verify that this is the current version before use.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Shell sizes are provided within parentheses for information and are not a part of the designator.
4. Dimensions apply after plating.
5. Normal mating key position shown. For alternate key positions see MIL-DTL-28840.
6. Assembly shall be made of a connector and backshell of the same material.

FIGURE 1. Dimensions and configuration – Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Classes: DJ, DJS, PJ, TJ and ZJ.

Insert arrangements: See the figure of MIL-STD-1698 corresponding to the desired shell size.

Contacts: Pin or socket, crimp type, styles P, S, D, E, F or G, in accordance with MIL-DTL-28840.

Shell polarization: 1 thru 6, key position.

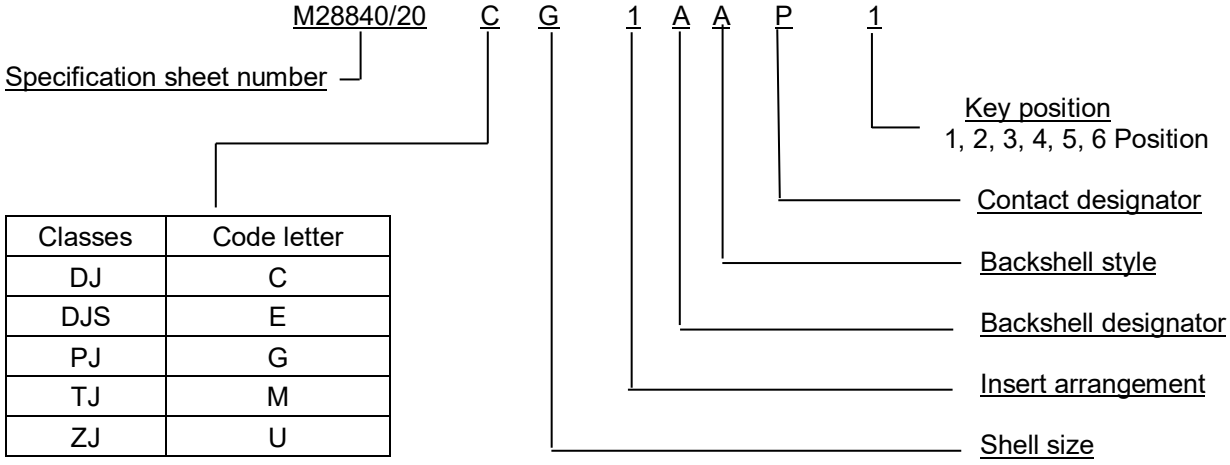
Marking: Part or Identifying Number (PIN) on shell.

Mating counterpart: Mating connectors are in accordance with MIL-DTL-28840/16, MIL-DTL-28840/17, MIL-DTL-28840/18, MIL-DTL-28840/19, MIL-DTL-28840/26, MIL-DTL-28840/28 and MIL-DTL-28840/29.

Installation and removal tool: In accordance with SAE-AS81969/33 and SAE-AS81969/34.

PIN's shown indicate aluminum alloy hardware. If corrosion resistant steel parts are required, the connector and accessory part numbers must be changed to designate corrosion resistant hardware.

PIN example:



Classes	Code letter
DJ	C
DJS	E
PJ	G
TJ	M
ZJ	U

RESTRICTIONS:

Classes DJ and DJS are inactive for NAVSEA new design, unless approved by NAVSEA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:41Z
Check the source to verify that this is the current version before use.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-28840, this document references the following:

MIL-DTL-28840/6
MIL-DTL-28840/10
MIL-DTL-28840/16
MIL-DTL-28840/17
MIL-DTL-28840/18
MIL-DTL-28840/19
MIL-DTL-28840/26
MIL-DTL-28840/28
MIL-DTL-28840/29
MIL-STD-1698
SAE-AS81969/33
SAE-AS81969/34

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2019-072)

Review activities:
Army – AV, MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:41Z
Check the source to verify that this is the current version before use.